THE 2-FIBRE PRODUCT OF A REPRESENTABLE MORPHISM WITH A SPACE-REPRESENTABLE CATEGORY IS REPRESENTABLE BY AN ALGEBRAIC SPACE

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Editorial note. No omission markers in-span. The two same-chapter inputs are not carded (one 'This is formal', one 'Omitted' hint) and are declared external.

Allowed standard prerequisites (editorial).

- Setoid-to-presheaf correspondence. Categories, lemmas setoid-fibres and 2-category-fibred-setoids (cross-chapter, named). Identify the 2-fibre product with S_F for a presheaf F and transfer the 1-morphism to a map F -> G.
- Representability transfer for setoid presheaves. lemma-map-fibred-setoids-representable-algebraic-spaces (same chapter, NOT carded, declared external). Transfers representability by algebraic spaces from the 1-morphism to the map F -> G.
- Bootstrap over an algebraic space. Bootstrap, lemma representable-by-spaces-over-space (cross-chapter, named). Concludes F is an algebraic space.
- Base-change stability. lemma-base-change-representable-by-spaces (same chapter, NOT carded, declared external). Makes the projection to Z representable by algebraic spaces.

Source reference. Original reference 02YS: focused-seeds/geometry/S9-alg-base-change-by-space-representable-by-space/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 04SC: focused-seeds/geometry/S9-alg-base-change-by-space-representable-by-space/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0045: focused-seeds/geometry/S9-alg-base-change-by-space-representable-by-space/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0302: sources/stacks-algebraic/pinned-original-source.tex, source lines 876–876. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has

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to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 04SY: sources/stacks-algebraic/pinned-original-source.tex, source lines 855–855. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme contained in Sch_{fppf}. Let $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ be categories fibred in groupoids over $(Sch/S)_{fppf}$ Let $f : \mathcal{X} \to \mathcal{Y}$, $g : \mathcal{Z} \to \mathcal{Y}$ be 1-morphisms. Assume*

(1) *f is representable by algebraic spaces, and*
(2) *$\mathcal{Z}$ is representable by an algebraic space over S.*

Then the 2-fibre product $\mathcal{Z} \times_{g,\mathcal{Y},f} \mathcal{X}$ is representable by an algebraic space.

Proof. This is a reformulation of Bootstrap, Lemma 02YS. First note that $\mathcal{Z} \times_{g,\mathcal{Y},f} \mathcal{X}$ is fibred in setoids over $(Sch/S)_{fppf}$. Hence it is equivalent to $\mathcal{S}_F$ for some presheaf F on $(Sch/S)_{fppf}$, see Categories, Lemma 0045. Moreover, let G be an algebraic space which represents $\mathcal{Z}$. The 1-morphism $\mathcal{Z} \times_{g,\mathcal{Y},f} \mathcal{X} \to \mathcal{Z}$ is representable by algebraic spaces by Lemma 0302. And $\mathcal{Z} \times_{g,\mathcal{Y},f} \mathcal{X} \to \mathcal{Z}$ corresponds to a morphism $F \to G$ by Categories, Lemma 04SC. Then $F \to G$ is representable by algebraic spaces by Lemma 04SY. Hence Bootstrap, Lemma 02YS implies that F is an algebraic space as desired. □